

Deep Research: Learning Without Retraining — Four Routes to Cheap, Local, Permanent Skill Acquisition

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TL;DR

Every route toward "updates so cheap and local that retraining stops mattering" works — up to the same wall. Test-time-learning layers write knowledge into fast weights during inference, but their inner objective is still co-trained by backprop in a slow outer pass: retraining is **moved, not eliminated**. Weight-free routes (retrieval, prototypes, skill libraries) genuinely need zero gradient updates, but only recombine structure someone else already paid for, and they fail exactly when new material demands a new *function* rather than a new *lookup*. Self-improvement from objective feedback is real and publishable but is bounded by the verifier's cost and by the base model's existing support. **The open, buildable gap: nobody has a long-horizon non-stationary benchmark that tests whether test-time learning actually delivers continual learning — and that is answerable on free compute.**

Executive Summary

The question posed — can a model acquire a skill by reading, practising, and integrating it, without a retraining phase — decomposes into four researched routes. This survey maps all four across vision and language, and reports where each one breaks.

Route 1 (cheap local weight updates) is the most mature and the most honest about its limits. A 2026 survey of the whole area states the consensus directly: parameter-efficient methods "remain vulnerable to catastrophic forgetting" [1]. Low rank reduces how much an update *can* damage, not whether it does. The null-space family (GPM, Adam-NSCL, SGP) has the cleanest theory — project new gradients away from stored input directions and old-task outputs are preserved to first order [2][3][4] — but is validated on Split-CIFAR-scale CNNs with ~100 stored directions, and the field already had to *soften* hard orthogonality to keep learning ability [3]. Architecture growth (Progressive Networks, MEGABLOCK, Expert Gateways) achieves zero forgetting by construction and pays with linear parameter growth [10][11][12]. Two verified results matter most here: LoRA does forget less than full fine-tuning, but only on *out-of-target-domain* tasks while also learning less [14]; and full fine-tuning buys +2% in-distribution while losing 7% out-of-distribution versus linear probing, with the damage caused by lower layers distorting pretrained features [15].

Route 2 (fast weights and test-time learning) is where the field actually moved in 2024-2026. Titans makes a memory module whose weights are updated *during inference* by online gradient descent driven by a surprise loss [20]; TTT layers make the hidden state itself a small trained model [21]; Nested Learning formalises the whole thing as optimisation problems running at different update frequencies [18]. This is the closest published architecture to the user's thesis. The verified caveats are decisive: Nested Learning's own abstract says continual learning is only "**potentially** unlocked" [18][19]; its continual-learning evidence is class-incremental text classification on CLINC/Banking77/DBpedia, not a long non-stationary stream [18]; every one of these layers has an inner objective, inner learning rate and chunk size learned by backprop-through-updates in a full pretraining

run; and the 2026 wave (LaCT, In-Place TTT, TTCD) is still calling itself "a promising step towards" continual learning [22][23][25][26], with the long-context branch now applying test-time updates to context utilisation itself [23]. **As of September 2026, no peer-reviewed long-horizon non-stationary benchmark validates test-time learning as continual learning** [27][28].

Route 3 (weight-free memory and skill libraries) delivers real, verified capability gains with zero weight updates: RETRO matches models 25× its size by retrieving from 2 trillion tokens [44]; Voyager discovers 3.3× more items and unlocks milestones up to 15.3× faster with no fine-tuning at all, storing skills as executable code [53]; FunSearch found genuinely new mathematics with no backprop whatsoever [72]. The boundary is precise and well-evidenced: retrieval supplies *recital*, not *skill* — models reliably answer questions whose evidence is in context yet fail to copy or transform that same content [47], and long-context accuracy is U-shaped in evidence position [48]. Knowledge lives as sparse circuits, so context can only *reactivate* what pretraining built [39]. Capacity is capped at ~2 bits of knowledge per parameter [41]. And the "5 dog photos" case works only because contrastive or supervised training already made dogs linearly separable — few-shot learning is a property of the representation, not of the learner [49][50][51].

Route 4 (self-improvement from objective feedback) works inside a narrow envelope: Absolute Zero, R-Zero and SWE-Gym all require a cheap, deterministic, ungameable verifier [57][58][59]. The theory explains why: self-improvement is *sharpening* — eliciting what the base model already produces with nonzero probability — not capability creation [63]. Model collapse is the structural failure [60], and the honest reading is that it may be unavoidable rather than merely engineering-fixable [61][62].

The convergence is the finding. All four routes hit a version of the same constraint: something outside the current update must already be there. Slow outer loop, pretrained structure, or an expensive verifier. What that leaves for a two-person, free-compute team is not a new architecture but the missing measurement — see §4.

1. Status Quo [Confidence: High]

1.1 Route 1 — cheap, local, interference-free weight updates

The field's own summary of this route is negative on its central promise. The 2026 survey of parameter-efficient continual fine-tuning organises the entire space into adapter-based, LoRA-based (orthogonal subspace, merging, masking), prompt-based, and selective families — and states that PEFT methods "remain vulnerable to catastrophic forgetting," while "full fine-tuning can result in catastrophic forgetting" [1]. Low rank is a *cost* mechanism, not an *interference* mechanism. That distinction is why an entire sub-literature exists to patch low-rank updates after the fact: orthogonalised LoRA factors [7], subspace-diversity regularisers to keep successive LoRA updates in distinct subspaces [7], dynamic rank allocation to fix LoRA's severe rank sensitivity in continual settings [6], task-aware dual branches [9], and hierarchical merging to collapse per-task adapters back into one [8].

Read that last one as the route's central trade-off stated plainly: either you keep N adapters — no forgetting, but linear module growth and no compositional transfer between tasks — or you merge into one, which is cheap but reintroduces interference [8]. PEFT's storage argument quietly cancels itself under continual use.

The theoretically cleanest family is null-space projection. Gradient Projection Memory stores a small basis (~100 directions) of input representations from a per-task core-set and projects new gradients onto its orthogonal complement, which provably preserves old-task function outputs to first order [2]. Adam-NSCL does the same in the null space of previous feature covariance, with an Adam-compatible implementation [4]; exact Orthogonal Gradient Descent predates both and is quadratic in task count, which is why everything after it approximates [5]. The mechanism is genuinely sound and genuinely limited in three ways at once: it preserves *activations on stored inputs*, not weights or behaviour off-manifold; the fixed-dimensional subspace eventually forces all new learning through the same handful of directions; and it was demonstrated on VGG-scale convolutional networks on sequential CIFAR, with no evidence at language-model scale [2][4]. The field's own correction is Scaled Gradient Projection, which abandons hard orthogonality for a soft, layer-wise-scaled projection precisely because the hard constraint is "unjustified" when gradients are strongly dependent — i.e. **plasticity had to be bought back by weakening the very constraint that prevented forgetting** [3].

Architecture growth is the only family with forgetting eliminated by construction rather than by penalty. Progressive Neural Networks add a column per task with lateral connections from all predecessors: zero interference, cost is parameters and inference time growing linearly with task count, plus a hard ceiling on how many tasks fit at deployment [10]. Capacity-limited pruning (PackNet) has the same ceiling in a single fixed network [13]. MEGABLOCK grows a residual-block stack per task [11]; Expert Gateways routes between per-task MoE experts and reported thousands of tasks without forgetting [12]; the 2026 successor line applies spectral structure to MoE routing for continual learning [16]. The concern these methods leave untested is stated in the survey literature as a limitation: growth "sacrific[es] scalability and inter-task generalizability" [1]. A per-task module plus a router is a **skill inventory addressed by lookup, not a skill integrated into a shared function**. Whether the system can *compose* two stored skills into a third behaviour is exactly the question a rigorous small-scale study could answer, and it is the difference between a library and a learner.

Two verified quantitative results anchor the critical view. Biderman et al. compared LoRA and full fine-tuning across programming and mathematics, in both instruction-tuning (~100K pairs) and continued-pretraining (20B tokens) regimes: in standard low-rank settings "LoRA substantially underperforms full finetuning," *nevertheless* it "better maintains the base model's performance on tasks outside the target domain" and "mitigates forgetting more than common regularization techniques such as weight decay and dropout," while full fine-tuning "learns perturbations with a rank that is 10-100X greater than typical LoRA configurations" [14]. Note the exact scope: LoRA's forgetting advantage is measured on *out-of-target-domain* tasks, not on the task it is learning. Kumar et al. supply the complementary warning across 10 distribution-shift benchmarks: full fine-tuning gains ~2% in-distribution but loses ~7% out-of-distribution relative to linear probing, because "the lower layers of the neural network change simultaneously and distort the pretrained features" — and their fix, LP-FT (learn the head first, then fine-tune), beats full fine-tuning by 1% ID and 10% OOD [15]. That is the strongest evidence in this route that **how** you update is more load-bearing than **how much** you update.

1.2 Route 2 — fast weights, test-time learning, nested learning

This route is the literal implementation of "cheap local parameter updates," and it is the fastest-moving area found in this survey. Titans is a neural long-term memory module whose weights are updated **during inference** by online gradient descent, driven by a surprise-based loss, with momentum and weight decay as learned components of the update; attention serves as short-term memory and a surprise gate decides what is worth writing [20]. TTT layers go further conceptually: "the key idea is to make the hidden state a machine learning model itself, and the update rule a step of self-supervised learning" [21]. Nested Learning generalises the pattern — any model is "a set of nested, multi-level, and/or parallel optimization problems, each of which with its own context flow," with Adam and SGD-with-momentum reinterpreted as associative-memory modules compressing gradient history, plus a Continuum Memory System and a self-referential module called Hope [18].

The lineage is worth stating because it deflates the novelty framing without dismissing the substance: fast weight programmers date to 1992 [31], Hinton co-authored fast-weights-in-static-computation in 2016 [32], linear transformers were shown to *be* fast weight programmers in 2021 [33], modern Hopfield networks give one-shot test-time retrieval as exponential-memory attention [34], and hypernetworks amortise the generation of task-specific weights [35]. Community criticism reads Nested Learning as a taxonomy of this existing line rather than a capability jump [36] — a Tier-3 source, but the same objection is available from the primary literature itself, since the flagship continual-learning claim is hedged with the word "potentially" and the abstract reports no quantitative results at all [18].

Attribution correction, because it will otherwise propagate: **Geoffrey Hinton is not an author of Titans, MIRAS, or Nested Learning** — those are Behrouz, Zhong, Razaviyayn and Mirrokni at Google Research [18] [20][24]. The frequently-cited 2024 Google post "Keeping Neural Networks Current with Test-Time Training" could not be located on arXiv, DBLP, or the implicated author's own publication list, and should not be cited [see Methodology]. The nearest real 2024 item is a Google post on reweighting *training* data under concept drift, which is a different mechanism [30].

What the route still requires is the slow loop. Every one of these layers has an inner objective, inner learning rate, chunk size and update rule that are themselves co-trained by backpropagation-through-updates during a full pretraining run [20][21][24]. The 2026 wave is explicit about this being unfinished business: LaCT's contribution is that prior TTT layers "operate with extremely low FLOPs utilization (often below 5%)" because they update fast weights every 16-64 tokens, and fixes it with 2K-1M-token chunk updates, nonlinear state up to 40% of parameters, and Muon for the online update, scaled to 14B-parameter video diffusion [22]; In-Place TTT ends by describing

itself as "a promising step towards a paradigm of continual learning in LLMs" while evaluating at 128k contexts [25]; TTCD performs inference-time "continual pre-training" over context length and beats DeltaNet, Gated DeltaNet, sliding-window attention and TTT [26]; a 2026 survey asserts that "test-time training extends CL from the training phase to inference" without introducing a benchmark that would test the claim [27]. TTT layers' own authors flagged that "TTT-MLP still faces challenges in memory I/O" [21].

And on the question that matters most for the user's goal — can it acquire a genuinely new skill after deployment with no outer phase? — the answer from this survey is that no published system demonstrates it, and the one 2026 result that pairs test-time training with a real non-stationary stream is video domain-incremental learning, not language [28].

1.3 Route 3 — weight-free memory, retrieval, prototypes, skill libraries

With weights frozen, three mechanisms exist. **Retrieve into context:** kNN-LM interpolates the LM's output with a nearest-neighbour lookup over cached context→next-token pairs, and improves perplexity and adapts to a new domain *by swapping the datastore, with zero gradient updates* [43]; RETRO retrieves from a 2-trillion-token corpus and matches the perplexity of models 25× its size (7.5B vs 175B, ~4% of the parameters) [44]; Memorizing Transformers extend effective context to ~262k tokens with non-differentiable kNN memory over past blocks [45]; RAG is the parametric/non-parametric template [46].

Read out with a cheap head: prototypical networks classify by nearest class mean with no negatives and no weight updates [49]; Bird & Gower formalise exactly when a frozen representation admits such a linear readout, i.e. few-shot learning is a property of the representation rather than of the learner [50]; CLIP zero-shot is the same operation precomputed as a dot product against a text embedding [51]. This is the honest answer to "show it five dogs and it knows what a dog is": that works, today, and it works because someone else already paid for the geometry that makes dogs separable from everything else. The remaining hard part is knowing when it *doesn't* know: open-set recognition shows that the reject rule is the difficult component, since thresholds calibrated on known classes degrade as the unknown gets semantically farther away [52].

Store skills outside the model: Voyager writes itself executable programs, verifies them in the environment, and accumulates an "ever-growing skill library of executable code for storing and retrieving complex behaviors," explicitly "bypass[ing] the need for model parameter fine-tuning" — 3.3× more unique items, 2.3× longer travel distance, key tech-tree milestones up to 15.3× faster than prior SOTA [53]. Reflexion stores verbal self-critiques in episodic memory and lifts HumanEval to roughly 91% pass@1 from ~80% for the same backend [68], with RLEF writing successful programs plus failing-test feedback into an in-context database [69]. FunSearch evolved programs producing new mathematical results with no backprop at all [72], Eureka evolves reward functions that beat hand-designed ones [73], and AlphaEvolve extended the recipe to open algorithmic problems [74].

The boundary is where the new material requires a **new function rather than a new lookup**. Retrieval raises answer availability, not competence: models reliably answer questions whose evidence is in context yet fail to copy or transform that same content [47], and multi-hop accuracy is U-shaped in evidence position, dropping from the mid-70s% to the mid-50s% in a 20-document setting [48]. Knowledge is instantiated as sparse circuits of information heads, relation heads and MLPs, so context can only reactivate circuits pretraining already built [39]. Capacity is hard-capped: models "can and only can store 2 bits of knowledge per parameter, even when quantized to int8" [41]. Most directly on point, the "knowledge onboarding" result is real but narrower than usually quoted — an optimally designed presentation format lets a model absorb roughly two orders of magnitude more knowledge than the least optimal one, behaving comparably to fine-tuning, **but only for knowledge composable from what the model already stores**; on inverse knowledge search the authors report performance "virtually 0%, regardless of the prompts" [42].

1.4 Route 4 — self-improvement from objective feedback

The strongest results all pay for a verifier. Absolute Zero has a single model propose code-reasoning tasks that maximise its own learning progress and solve them, with reward "derived from a code executor" — no human data, no external reward model [57]. R-Zero runs Challenger/Solver co-evolution from one base model and reports +6.49 on math-reasoning and +7.54 on general-domain benchmarks for Qwen3-4B-Base, but substitutes *consensus agreement* for ground truth, i.e. an internal verifier [58]. SWE-Gym keeps only issues with reproducible sandbox test runs, and self-training on that verifiable subset yields up to 19% absolute gains in resolve rate, reaching 32.0% on SWE-bench Verified [59]. FunSearch and AlphaEvolve make the same point in search form: the evaluator is the entire epistemics, and it must be fast, cheap, deterministic and ungameable [72][74].

The self-taught-reasoner lineage supplies the honest effect sizes. STaR bootstraps its own rationales but filters them against a small set of ground-truth answers, so the loop was never actually label-free [65]; Quiet-STaR removed that dependency and beat prior self-teaching methods by a large margin in token efficiency, while the downstream accuracy gains remained small in absolute terms [64]. Self-teaching reliably improves *efficiency and format*, weakly improves *correctness*, and is essentially silent on acquiring a genuinely new skill.

The theory of why there is a ceiling: self-improvement decomposes into generation and verification, and works by *sharpening* — concentrating probability mass on answers the base model already produces with nonzero probability — so gains are bounded by the pretraining distribution and by verifier accuracy, and the loop stalls when task success probability is near zero [63]. SPIN proves its own limit: the self-play fixed point is the SFT data distribution, so it converges *toward* existing human data, never beyond it [71]. Self-rewarding with an LLM-as-judge improves AlpacaEval 2.0 from ~28.5% to ~40.4% over three iterations, but the gains become unattributable to correctness — judge bias is the reward-hacking channel [70]. And model collapse is the structural risk: models trained on their own outputs "progressively lose knowledge of the tails," which is exactly what an agent needs to notice its own gaps [60]; accumulating real plus synthetic data preserves the bulk but not the tails and needs unbounded replay [61]; the bleakest published reading is that collapse "is a statistical phenomenon and may be unavoidable" [62].

One widely-repeated claim in this area did **not** survive checking and is corrected here: ReST^EM does *not* show that one round of self-training plus model merging beats many iterations. Its text says "using multiple iterations of ReST^EM result in larger performance gains" and "MATH benefits from performing multiple iterations"; the real limitation is task-specific — "most of the gains for APPS come from the first iteration, and more iterations lead to a regression on both APPS and HumanEval," indicating overfitting to a small problem set [67]. Diminishing returns, yes. Recursion-is-poison, no.

2. Emerging Trends [Confidence: High]

Test-time learning became the mainstream research answer to continual learning. Between December 2024 and August 2026 the Titans → MIRAS → Nested Learning → LaCT / In-Place TTT / TTCD line moved from "a new memory module" to a family of architectures with an explicit continual-learning narrative [20] [24][18][22][25][26]. Two 2026 surveys now describe the shift as settled direction — "test-time training extends CL from the training phase to inference" [27] — while the empirical validation lags the rhetoric [28]. That gap between confident framing and absent long-horizon evidence is the single most exploitable finding in this report.

Skill libraries are the practitioner's answer to no-retraining. Voyager's recipe (automatic curriculum → write code → environment feedback → verified program → embedding-indexed library → compose) is now the template for a visible 2026 line such as AutoSkill, "experience-driven lifelong learning via skill self-evolution" [75], and self-play software training [76]. This is the mechanism closest to "reads documentation, becomes able to do the thing," and it works because the artefact is *executable and self-verifying*, not because the model understood anything.

The PEFT-for-continual-learning literature is converging on subspace discipline. Orthogonal LoRA factors, subspace-diversity regularisation, dynamic rank allocation, hierarchical merging, dual-branch LoRA and spectral MoE all share one premise: keep successive updates in different subspaces [1][6][7][8][9][16]. That is null-space thinking (Route 1's theory family) reappearing inside LoRA engineering — the two halves of Route 1 are quietly merging.

Verifier economics is becoming the bottleneck everyone names. Every positive self-improvement result is now reported alongside its verification cost — thousands of code-execution rollouts per prompt [57], sandboxed reproducible environments [59], millions of evaluator calls [72][74]. Practitioner folklore puts the useful depth of a self-improvement loop at "about 3 rounds" [76a]. The frontier question has shifted from *can* models self-improve to *what does it cost to know when they have*.

Compositionality is the emerging test of whether stored skills are real. The reversal curse showed that gradient-trained knowledge does not generalise to logical inverses: fine-tuned on "A is B," GPT-3 forward accuracy was 96.7% while reverse was 0.1%, robust across model sizes and families and "not alleviated by data augmentation," with real celebrity data at 79% forward versus 33% reverse for GPT-4 [54]. Crucially, the same paper concedes that when "A is B" appears *in-context*, models can deduce the reverse — so this is a finding about

trained knowledge, not a bound on documentation-based acquisition [54]. A mitigation literature now exists [55] [56]. For anyone building a skill library, the reversal test is a free, sharp probe of whether stored skills are understood or merely addressed.

3. Critical Assessment [Confidence: High]

3.1 The wall all four routes hit

Strip the mechanisms apart and one constraint recurs in three costumes.

Costume 1 — the slow outer loop. Fast weights update at inference, but the inner objective, learning rate and update rule are learned by backprop during pretraining [18][20][21][24]. Nested Learning's own hedge is "potentially" [18]. Retraining is relocated into the design of the inner loop.

Costume 2 — borrowed structure. Prototypes, linear probes and kNN memory need a representation in which the target is already separable [49][50][51][43]. RETRO's win belongs as much to the retriever's encoder as to the frozen model [44]. Knowledge circuits mean context can only reactivate what pretraining built [39]. A baby gets its pre-structure from evolution plus years of sensorimotor experience; a model gets it from someone's training run. The honest form of the question is not "can we eliminate training" but "**how much pre-structure must exist before few-shot acquisition starts working?**" — and that is measurable.

Costume 3 — verifier economics. Self-improvement from feedback is real only where an external, cheap, ungameable oracle exists [57][59][72], and it cannot exceed the base model's support [63][71]. The bill is paid by the evaluator, not for free.

3.2 Where "cheap and local" is a false promise

Local in parameter space is not local in function space. Full fine-tuning's damage is that lower layers "change simultaneously and distort the pretrained features" — an update that is small in norm can be catastrophic in effect [15]. LoRA's rank-10-100×-smaller perturbation is precisely what buys its forgetting resistance [14], but the same restriction is why it "substantially underperforms full finetuning" on the target domain [14] — the cheapness is the limitation. Null-space constraints trade plasticity for stability explicitly [3]. And per-task module growth avoids interference by never sharing anything, which forfeits the composition that made the network worth using in the first place [10][12].

3.3 The measurement hole (this is the finding)

Three specific absences, each verified in this pass:

1. **No long-horizon non-stationary benchmark for test-time learning.** Nested Learning's continual evidence is CLINC/Banking77/DBpedia class-incremental text classification [18]; 2026 TTT papers evaluate at 128k-1M token *contexts*, not drifting *tasks* [22][25][26]; the survey literature asserts the connection without a benchmark [27]; the one 2026 stream result is video domain-incremental [28].
2. **No test of compositional integration in skill libraries.** Voyager stores and retrieves; no paper surfaced here demonstrates that composed skills generalise to a neighbouring problem, and the acknowledged failure mode is silent-but-wrong programs entering the library because they executed [53].
3. **No clean separation of "acquired" from "elicited" in self-improvement.** The sharpening theory predicts the confound [63]; RANK was built to isolate exactly this and its numbers could not be verified from accessible sources [40].

3.4 What a skeptic should say, and what the data says back

"You cannot learn without changing weights." — Correct so far: no published system acquires a genuinely new function post-deployment with zero parameter change, and Route 3's successes are recombination within pretrained structure [39][41][50]. *"Retrieval is enough — just give it a bigger window."* — Contradicted: in-context benefit provably decays as context grows and reasoning depth increases, which is why tool-use and external memory dominate unaided context [29][47][48]. *"Self-improvement will bootstrap the rest."* — Contradicted

twice: sharpening bounds it [63], and collapse tails it [60][62]. "*So the goal is impossible.*" — Not what the evidence says. It says the goal is *bounded*, and the bound is located: pre-structure, outer loop, verifier. Those are engineering surfaces, not walls.

4. Gap Analysis — What MK and Aadit Can Actually Do [Confidence: Medium]

Constraints: two people, new to research practice, free cloud compute (Colab/Kaggle), one collaborator with a 6GB RTX 3060 and a pre-registered GOCO protocol, and an end goal of self-directed acquisition without retraining. Ranked by feasibility $\times$ novelty:

- [] **Gap 1 (strongest fit): build the missing long-horizon benchmark.** Take an existing test-time-learning implementation at small scale (a TTT layer or a Titans-style memory module on a 10-50M parameter model, or even the associative-memory toy from [34]) and evaluate it on a *non-stationary stream of tasks over time*, not a long context — measuring worst-case accuracy during the stream (the stability-gap protocol from the companion survey), drift, and how much the fast-weight state actually retains. Nothing in the 2024-2026 literature does this [18][22][25][26][27][28]. It is a benchmark paper, it is exactly the hole §3.1 identifies, and it needs no large training run.
- [] **Gap 2: the compositionality probe for skill libraries.** Implement Voyager-style accumulation on a *verifiable* toy domain (Aadit's GOCO compiler is ideal — objective, deterministic, cheap to run) and ask the question nobody answers: does a stored skill transfer when the task is inverted, rephrased, or composed? Use the reversal-curse protocol as the test [54]. This is directly publishable, uses existing infrastructure, and is the sharpest available attack on "recital vs skill" [47].
- [] **Gap 3: how much pre-structure does few-shot need?** Systematically degrade or restrict a frozen representation (layer choice, width, training-data diversity) and measure where 5-shot prototype classification collapses [49][50][51]. This converts the vague "someone else already trained it" objection into a curve, and curves get cited.
- [] **Gap 4: update-order and head-first protocols at CL scale.** LP-FT's finding — learn the head before touching features, +10% OOD — is a free, protocol-level intervention [15]. Test whether it holds under class-incremental evaluation and whether it changes the stability-plasticity frontier. Cheap, rigorous, immediately useful to the CL community.
- [] **Do not attempt:** training a new memory architecture (Route 2's cost is the outer pretraining run), a self-improvement loop without a cheap deterministic verifier, or anything requiring many-round recursion on self-generated data [60][62].

Positioning note. Gap 1 and Gap 2 are complementary to Aadit's GOCO protocol rather than competing: he owns a pre-registered instrument with sealed suites and a NO-GO result at Phase 1; these gaps need exactly that discipline and supply it a literature anchor. His Phase 1 floor effect (frozen model, full trusted docs, 0/24) is a data point on the same boundary RANK and the onboarding results describe [40][42] — worth reporting together rather than as an anomaly.

5. Action Plan

- [] **Week 1-2:** read the four anchor papers properly — PEFT-CL survey [1], Titans [20], TTT layers [21], Voyager [53] — and write one page each on *what would make this claim wrong*.
- [] **Week 2:** re-verify the three claims this pass could not confirm before building on them: RANK's numbers [40], Absolute Zero's headline gain [57], and the "3 rounds" practitioner claim [76a].
- [] **Week 3-6:** reproduce a small test-time-learning layer on a ≤ 50 M model; get one non-stationary stream running end to end with worst-case-during-training evaluation.
- [] **Week 6:** decide Gap 1 vs Gap 2 based on which instrument is actually working; do not run both.
- [] **Ongoing:** publish a short public note per experiment. Report verifier cost (rollouts, wall-clock) alongside accuracy in every self-improvement-adjacent run — it is the field's missing number [57][59][72].
- [] **Coordination:** send Aadit §3.1 and the reversal-curse caveat [54]; his Phase 1R protocol should account for in-context vs trained knowledge separately.

- [] **Honesty rule for the eventual paper:** never write "continual learning" where the evidence is "long context," and never write "learns" where the mechanism is "retrieves."

6. Open Questions & Caveats

1. **Does test-time learning actually deliver continual learning?** Unresolved. The literature's own word is "potentially" [18], and no benchmark exists (§3.1.1). This is a gap, not an answer.
2. **RANK numbers unavailable.** The paper exists with Shah as first author on OpenReview; its venue attribution and all quantitative claims are unconfirmed because OpenReview blocked automated access and no arXiv preprint exists [40]. Treat as qualitative only.
3. **Absolute Zero's ~20-point gain unverified.** Supported qualitatively (SOTA among zero-human-data models); the number and base-model size were not retrievable and are stated here only as "double-digit average gains over a same-size model trained without self-play" [57].
4. **MEGABLOCK / Expert Gateways / several 2026 items were located but not read** [11][12][16][17][26][27][28][37][38][75][76]. Their claims are carried at title/abstract level.
5. **Four citations were fabricated by the retrieval agents and removed** — see Methodology. Two arguments that would have been load-bearing (that parameter-space constraints measurably hurt future learning; that SGD already concentrates gradients into a low-dimensional subspace) are therefore **unsourced** in this report and are stated as open questions rather than findings.
6. **Scope not covered:** spiking/neuromorphic routes (in the companion survey), RL-based continual learning, and any German/Chinese-language literature. Vision coverage is thinner than language by design of the routes.
7. **The strongest counter-argument to the whole programme** — that "cheap local updates" is a re-labelling exercise because the cost always reappears as pretraining, datastore, or verifier compute — is well supported by this evidence and is not resolved by anything here.

Methodology

Depth: **deep**. Pipeline: Phase 0 clarify (3 questions, one round) → Phase 1 scope (4 routes × 2 domains, ~40 search terms incl. critical-view terms) → Phase 2 Wave 1 (4 parallel retrieval subagents, ~60 candidate sources) → Phase 3 triangulation → Wave 2 gap-fill (3 targeted subagents, indices 90/100/110) → Phase 3.1 citation verification (1 subagent, 6 highest-impact claims) → Phase 3.5/4 outline refinement and red-team → Phase 5 write. Total: 8 subagents. Source list trimmed to 76 after deduplication (Voyager, FunSearch, Reflexion, ReST and Kumar et al. each surfaced from two agents and were merged).

Citation integrity — the most important methodological result. The verification waves found **four citations that do not exist**, all produced by retrieval agents stating them with confident authors and venues: 1. "Is Future Learning Affected by Parameter-Space Constraints for Previous Tasks?" (attributed Everaert et al., NeurIPS 2021) — absent from NeurIPS 2021/2022/2023 proceedings, arXiv, OpenReview and Crossref. 2. "Gradient Descent Performs Projections onto Orthogonal Gradient Subspaces" (attributed Ghorbani et al., NeurIPS 2021) — absent from all indexes. 3. "Fine-Tuning, LoRA and Stochastic Freezing Differentially Affect LLM Robustness" (attributed Kumar et al., NeurIPS 2023) — absent; the real Kumar et al. paper does not study LoRA or LLMs. 4. "Keeping Neural Networks Current with Test-Time Training" (attributed Google/Alvarez-Melis/Hinton, 2024) — absent from arXiv, DBLP and the implicated author's own publication list; the nearest real 2024 item concerns training-data reweighting under drift [30]. Consequences applied: two arguments were **removed from the critical assessment** rather than kept with weaker sourcing, and are now listed as open questions (§6.5).

Six corrections applied from verification: RETRO's arXiv ID 2112.09114 is a top-quark physics paper — correct ID 2112.04426, and "~2% of parameters" is wrong (25× fewer ≈ 4%) [44]; the FunSearch arXiv ID 2210.12726 is a graph-theory paper — cite Nature 625(7995):468-475, online 14 Dec 2023 [72]; the reversal-curse title/venue were wrong — it is "The Reversal Curse," ICLR 2024, arXiv 2309.12288, and its in-context caveat was added [54]; "Physics of LM Part 3.2" is titled *Knowledge Manipulation* and its ~650× figure compares presentation formats, not onboarding-vs-pretraining [42]; ReST is Gulcehre et al., not Casper et al. [66]; ReST^EM does **not** claim one round plus merging beats many — its text says multiple iterations give larger gains, with task-specific regression after the first on code [67]; Kumar et al.'s correct ID is 2202.10054, and the arXiv ID given in the brief (2106.09685) is the LoRA paper [15].

Deviation from the skill: Phase 5 synthesis was performed directly by the lead agent rather than by a Synthesis subagent, because all four route takeaways and the verification tables were already in working context; no external retrieval was needed for the write-up. Outline change versus the Phase 1 plan was under 50% (added §3.1 "the wall" and §3.3 "measurement hole" as evidence-driven sections), so no user checkpoint was required.

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Source Extracts

Condensed to the data actually used above. Full extracts are recoverable from the retrieval agents' outputs on request.

Route 1 — cheap local weight updates

- **[1] PEFT-CL survey (Neurocomputing 2026)** — taxonomy of adapter/LoRA/prompt/selective CL methods. *Key quote:* PEFT methods "remain vulnerable to catastrophic forgetting"; architecture growth "potentially sacrific[es] scalability and inter-task generalizability." Type: academic. Tier 1.
- **[2] GPM (ICLR 2021)** — projects gradients onto the orthogonal complement of a ~ 100 -direction core-set subspace; first-order preservation of old-task outputs; Split-CIFAR/CIFAR100-B0 scale. Tier 1.
- **[3] SGP (AAAI 2023)** — hard orthogonalisation is "unjustified"; soft layer-wise scaling needed $\rightarrow$ the plasticity cost is real and acknowledged. Tier 1.
- **[6] PEARL (TMLR 2026)** — "low-rank adapters ... are highly sensitive to rank selection"; rank choice, not low rank, drives reported interference resistance. Tier 1.
- **[10] Progressive Networks (2016)** — zero forgetting by construction; parameters and inference grow linearly per task. Type: academic. Tier 1 [foundational].
- **[14] LoRA Learns Less and Forgets Less (TMLR 2024)** — verified abstract: LoRA "substantially underperforms full finetuning" in standard low-rank settings; "better maintains the base model's performance on tasks outside the target domain"; "mitigates forgetting more than ... weight decay and dropout"; full FT perturbation rank 10-100 $\times$ larger. Type: academic. Tier 1.
- **[15] Kumar et al. (ICLR 2022 Oral)** — verified: FT +2% ID / -7% OOD vs linear probing across 10 shift benchmarks; distortion of lower layers; LP-FT +1% ID / +10% OOD. Type: academic. Tier 1.

Route 2 — fast weights, test-time learning, nested learning

- **[18] Nested Learning (NeurIPS 2025)** — verified author list (Behrouz, Razaviyayn, Zhong, Mirrokni; **no Hinton**); nested optimisation framing; Adam/SGD as associative memory; Continuum Memory System; HOPE introduced in §8-§9. *Key hedge:* "potentially unlocking effective continual learning capabilities"; CL evaluation = CLINC (150 intents/10 domains), Banking77, DBpedia; no long-horizon/non-stationary stream. Tier 1.
- **[20] Titans (NeurIPS 2025)** — verified: memory weights updated during inference by online gradient descent with surprise loss, momentum and weight decay; ">2M context window"; higher needle-in-haystack accuracy. HOPE does **not** appear in Titans v1. Tier 1.
- **[21] TTT layers (ICML 2025)** — verified: "make the hidden state a machine learning model itself, and the update rule a step of self-supervised learning"; 125M-1.3B params; Mamba cannot keep reducing perplexity past 16k context while TTT can; author limitation "TTT-MLP still faces challenges in memory I/O." Tier 1.
- **[22] Test-Time Training Done Right / LaCT (ICLR 2026 poster)** — verified: prior TTT layers run "extremely low FLOPs utilization (often below 5%)" updating every 16-64 tokens; LaCT uses 2K-1M-token chunks, state up to 40% of params, Muon online updates, 14B video diffusion at 56K tokens. Its criticism is compute inefficiency, **not** under-tuned baselines. Tier 1.
- **[24] MIRAS (2025)** — framework of four memory design choices; models Moneta, Yaad, Memora; explicit scaffolding for Nested Learning. Tier 1.
- **[25][26][27][28] 2026 TTT wave** — In-Place TTT "a promising step towards a paradigm of continual learning in LLMs" (128k contexts); TTCD beats DeltaNet/GDN/SWA/TTT via inference-time continual pre-training; survey claims TTT "extends CL from the training phase to inference" with no new benchmark; only real-stream result is video domain-incremental. Tier 1.
- **[36] Reddit critique of Nested Learning** — existence and shape of the "rebranding" objection only. Type: community. Tier 3.

Route 3 — weight-free memory, prototypes, skill libraries

- **[39] Knowledge Circuits (NeurIPS 2024)** — knowledge as sparse subnetworks (information + relation heads + MLPs); context reactivates existing circuits; editing damages circuits and general ability. Tier 1.

- **[41] Physics 3.3 (2024)** — verified verbatim: models "can and only can store 2 bits of knowledge per parameter, even when quantized to int8"; $7B \approx 14B$ bits, exceeding English Wikipedia + textbooks combined. Scope: factual knowledge as tuples. Tier 1.
- **[42] Physics 3.2 Knowledge Manipulation (ICLR 2025)** — corrected: $\sim 650\times$ compares optimal vs least-optimal *presentation format*, not onboarding vs pretraining; onboarding works only for composable knowledge; inverse knowledge search "virtually 0%, regardless of the prompts." Tier 1.
- **[43][44][45][46] kNN-LM / RETRO / Memorizing Transformers / RAG** — datastore swap adapts with zero gradient updates; RETRO 7.5B matches $25\times$ -larger models with a 2T-token datastore ($\approx 4\%$ of parameters, corrected); $\sim 262k$ effective context; RAG parametric+non-parametric template. Tier 1.
- **[47][48] Where is my context? / Lost in the Middle** — models answer from in-context evidence yet fail to copy or transform it; multi-hop accuracy U-shaped in position, mid-70s% $\rightarrow$ mid-50s% (20-doc, GPT-3.5-turbo setting; exact figures to verify). Tier 1.
- **[49][50][51] Prototypical Networks / Bird & Gower / CLIP** — nearest-class-mean with no negatives and no updates; linear readout exists only when the representation makes classes separable; CLIP zero-shot is a precomputed dot product. The 5-dog case works because prior training paid for the geometry. Tier 1.
- **[53] Voyager (TMLR 2024)** — verified verbatim from abstract: "3.3x more unique items, travels 2.3x longer distances, and unlocks key tech tree milestones up to 15.3x faster"; "bypasses the need for model parameter fine-tuning"; ever-growing executable skill library. Caveat: depends on GPT-4's pre-existing JavaScript/Minecraft knowledge and a cheap deterministic environment. Tier 1.
- **[54] Reversal Curse (ICLR 2024)** — verified: forward $96.7 \pm 1.2\%$ vs reverse $0.1 \pm 0.1\%$ (GPT-3-175B); $50.0 \pm 2.1\%$ vs $0.0 \pm 0.0\%$ reverse; Llama family $>80\%$ vs $<7\%$; GPT-4 real-celebrity 79% vs 33% ($n=1,573$, pass@10); "not alleviated by data augmentation"; scaling plots flat. **Critical caveat:** in-context presentation lets models deduce the reverse. Mitigations [55][56]. Tier 1.

Route 4 — self-improvement from feedback

- **[57] Absolute Zero (NeurIPS 2025)** — verified qualitative: self-play task proposal + solving, reward "derived from a code executor," SOTA among zero-human-data models, outperforming zero-setting models using tens of thousands of curated examples. Headline point-gain **unverified**. Tier 1.
- **[58][59] R-Zero / SWE-Gym** — +6.49 math and +7.54 general reasoning on Qwen3-4B-Base with consensus-as-verifier (the risky configuration); SWE-Gym 2,481 reproducible issues, up to 19% absolute resolve-rate gains, 32.0% SWE-bench Verified. Tier 1.
- **[60][61][62] Model collapse line** — Nature 631:755-760; "progressively lose knowledge of the tails"; accumulation preserves bulk not tails and needs unbounded replay; Borji's note: "a statistical phenomenon and may be unavoidable." Tier 1.
- **[63][71] Sharpening / SPIN** — self-improvement = generation + verification, concentrating mass on already-supported answers; requires nonzero success probability and an accurate verifier; SPIN's fixed point is the SFT distribution. Tier 1.
- **[66][67] ReST / ReST^EM** — corrected authorship (Gulcehre et al., DeepMind; machine-translation experiments; no venue recorded); ReST^EM (TMLR 2024) says "multiple iterations ... result in larger performance gains," with APPS/HumanEval regression after the first iteration — the "one round + merging beats many" claim is **not in the paper**. Tier 1.
- **[68][69][70] Reflexion / RLEF / Self-Rewarding** — verbal feedback into episodic memory, $\sim 91\%$ vs $\sim 80\%$ HumanEval pass@1 (verify exact); RLEF writes successes plus failing tests into context; self-rewarding AlpacaEval2 $\sim 28.5\% \rightarrow \sim 40.4\%$ over 3 iterations with judge-bias confound. Tier 1.
- **[72][73][74][75][76] Search-as-learning** — FunSearch (Nature 625:468-475, online 14 Dec 2023) new cap-set/bin-packing results with no backprop; Eureka evolves reward functions; AlphaEvolve improves matrix multiplication $\sim 0.5\%$ and GPU kernels, needing thousands of evaluator calls; 2026 successors AutoSkill and self-play SWE-RL. All require a fast, cheap, ungameable verifier. Tier 1/2.